

# Sparse Reconstruction Algorithm for Nonhomogeneous Counting Rate Estimation

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**Abstract**—One of the main objectives of nuclear spectroscopy is the estimation of the counting rate of unknown radioactive sources. Recently we proposed an algorithm based on a sparse reconstruction of the time signal in order to estimate precisely this counting rate, under the assumption that it remained constant over time. Computable bounds were obtained to quantify the performances. This approach, based on a post-processed approach of a nonnegative sparse regression of the time signal, performed well even when the activity of the source was high. The purpose of this paper is to present an extension of the previous method for an activity varying over time. It relies on the same preliminary sparse reconstruction. However the post-processed and plug-in steps are made differently to fit the nonhomogeneous framework. The adapted bounds are presented, and results on simulations illustrate the advantages and limitations of this method.

**Index Terms**—Statistical signal processing, nonparametrics, sparse signal representation, LASSO, nuclear spectroscopy, activity estimation.

## I. INTRODUCTION

**R**ANDOM point processes are extensively used in numerous fields to model localized events. One of the most common models is the Poisson process, homogeneous or nonhomogeneous. Though simple, its extensive use stems from its relevance in modeling numerous phenomena occurring in practical situations. Typical applications include, in particular, nuclear spectroscopy [1], [2], PET imaging [3], biomedical engineering and neural spike modeling [4]–[7], optics, communications and more [8].

In this paper we present an application stemming from the nuclear science framework, namely the estimation of a time-varying radioactive activity of a gamma emitting source. In a typical spectroscopy experiment, an unknown radioactive source emits photons at random times, which are modeled by a Poisson process. Each photon which interacts with a semiconductor detector produces electron-hole pairs whose migration generates an electrical pulse of finite duration. We can therefore estimate the activity of the source by counting the number of activity periods of the detector. We refer the reader to [9] for further insights on the physical aspects in this framework. However, when the source is highly radioactive, the durations of the electrical pulses may be longer than their interarrival times, thus the pulses can overlap. In gamma spectrometry, this phenomenon is referred to as pileup. Such

a distortion induces an underestimation of the activity, which become more severe as the counting rate increases. A typical recorded signal at very high activity is displayed in Figure 1a, with a zoom-in on a segment showing numerous pileups in Figure 1b.

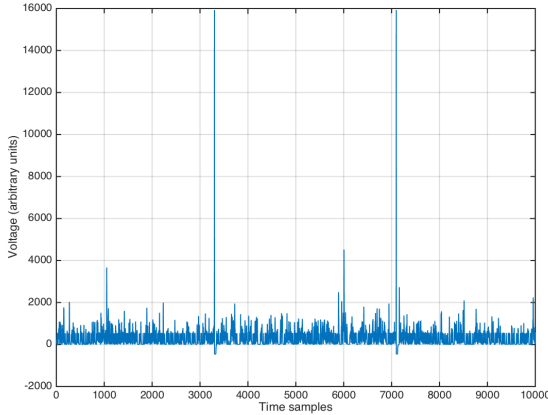
The problem to address is related to the estimation of the intensity of shot-noise processes from one or more sample paths. Numerous methods have been suggested in that matter: [10]–[12] used a variant of the EM algorithm, considering the intensity estimation in a missing data framework, while sequential Monte-Carlo methods have been used in [13], [14]. Uses of both parametric [15] and nonparametric [16]–[18] models have also been investigated. This paper can be related to [19], in which the intensity of a nonhomogeneous Poisson process is estimated from sampled data of a shot-noise process with known impulse response, and to [20], in which the joint estimation of the intensity and the impulse response is performed. Both rely on a penalized regression of the observed signal with the Tikhonov regularization. Though the approach proposed in this paper shares a similar basis, we note that [19], [20] rely on a time-invariant impulse response from the shot-noise model. Such assumption does not hold in our framework, since the pulse shape associated with an individual photon may vary. Recent advances in compressive sensing and sparse reconstruction also allowed reinvestigation of this problem with a new outlook [21]–[24]. Using a coarse dictionary of possible shapes and sparse reconstruction methods, [25] proposed a two-step algorithm to estimate the intensity of a homogeneous Poisson process from a sample path, and derived practical bounds to quantify the performances of the algorithm.

The present paper is a generalization of [25], [26] for nonhomogeneous Poisson processes (NHPPs). It is organized as follows: in the next section we present the model and assumptions used for the estimation of an NHPP's intensity, given a recorded signal stemming from a spectrometrical apparatus. We then provide practical, computable bounds justifying the proposed algorithm for the estimation of the arrival times. The latter is done by means of a novel iterative algorithm, based on a LASSO-driven selection of sampling points of the signal as potential starting points of electrical pulses. The obtained estimates of the arrival times are then plugged in into a non-parametric estimator of the intensity whose error decomposition is provided. Eventually, results on simulations show the efficiency of the proposed approach. Results obtained on real datasets also illustrate the efficiency of the arrival times estimation procedure.

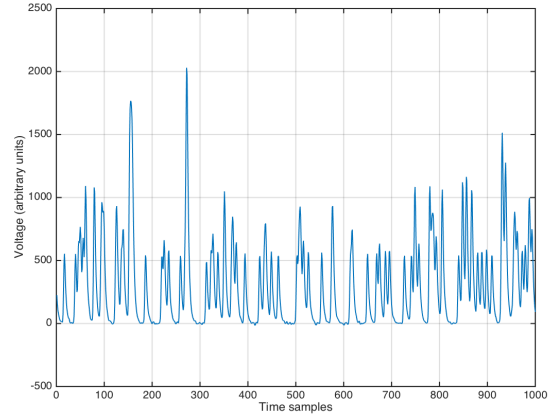
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(a) Example of a spectrometric signal - the highest values correspond to detector's overshoots.



(b) Example of a spectrometric signal exhibiting numerous pileups.

Fig. 1. Typical signals resulting from a nuclear spectroscopy experiment.

## II. OVERVIEW OF THE ALGORITHM

Alongside the paper, we will use a model similar to the one in [25]. We will briefly recall it for convenience, and present the proposed method for activity estimation in a nonhomogeneous setting.

### A. Model Description

We observe a signal on the time interval  $[0; T]$  uniformly sampled on  $\mathcal{T} \triangleq \{0 = t_0, t_1, t_2, \dots, t_{N-1}, t_N = T\}$  with sampling period  $\Delta t$ ,

$$y_i = \sum_{n=1}^M E_n \Phi_n(t_i - T_n) + \varepsilon_i, \quad 0 \leq i \leq N-1, \quad (1)$$

where  $\{T_n, 1 \leq n \leq M\}$  is the sample path of a non-homogeneous Poisson process (NHPP) with unknown, bounded, intensity  $\lambda(t)$ ,  $\{E_n, 1 \leq n \leq M\}$  is a sequence of independent and identically distributed (iid) random variables representing the photonic energies with unknown probability density function,  $\{\Phi_n, 1 \leq n \leq M\}$  is a sequence of functions to be defined later which characterize the electric pulse shapes generated by the photons, and  $\{\varepsilon_i, 0 \leq i \leq N-1\}$  is a sequence of iid Gaussian random variables with zero mean and variance  $\sigma^2$  representing the additional noise of the input signal. We assume in this paper that the activity  $\lambda(t)$  of the underlying NHPP is upper bounded, with finite  $\ell_\infty$ -norm  $\|\lambda\|_\infty$ , and that the energies  $E_n$  are almost surely bounded by positive constants  $e_{\min} \leq E_n \leq e_{\max}$ .

As noted in [25], neither the energies nor the shapes are known in practice. However, mild and plausible assumptions can be made: most HPG detectors deliver pulses characterized by rapid growth created by the charge collection followed by exponential decay as the charges migrate to the detector electrodes. Thus, a natural idea is to rely on some user predefined dictionary of truncated gamma shapes in order to obtain a reliable model of (1) that we can work with. Since a gamma shape is parametrized by two parameters — its shape

and scale —, we define a set of  $p$  pairs of such parameters by  $\theta \triangleq \{(\theta_1^{(s)}, \theta_2^{(s)}); s = 1, 2, \dots, p\}$ . For all  $s = 1, \dots, p$ , we define the following pulse shape:

$$\Gamma_s(t) \triangleq c_s t^{\theta_1^{(s)}} \exp(-\theta_2^{(s)} t) \mathbf{1}(0 < t \leq \tau \Delta t), \quad (2)$$

where  $\tau$  is a positive constant integer defining the common support of the pulse shapes, and  $c_s$  is a normalizing constant chosen so that  $\frac{1}{N} \sum_{i=0}^{N-1} \Gamma_s(t_i)^2 = 1$ . This normalization allows to directly relate the photon energies  $E_n$  to the regressors to be defined later. It also simplifies the proofs of the presented results. Accordingly, we define the following  $N \times p$  matrix  $\mathbf{A}_k$  whose columns are sampled versions of the previously defined pulse shapes, translated by  $t_k$ :

$$\mathbf{A}_k \triangleq [\Gamma_s(t_i - t_k)]_{0 \leq i \leq N-1, 1 \leq s \leq p}. \quad (3)$$

We further on refer to  $\mathbf{A}_k$  in (3) as the *time block* associated to the  $k$ -th point. We now define a global dictionary  $\mathbf{A}$  by concatenating these time blocks:

$$\mathbf{A} \triangleq [\mathbf{A}_0 \quad \mathbf{A}_1 \quad \dots \quad \mathbf{A}_{N-1}]. \quad (4)$$

Given any vector  $\beta$  of size  $pN$ , it is thus naturally decomposed along the timeblocks:  $\beta = [\beta_0^T; \dots; \beta_{N-1}^T]^T$ , where for all  $i$ ,  $\beta_i \in \mathbb{R}^p$ ; its block pattern can be introduced as:

$$J(\beta) \triangleq \{i; \beta_i \neq 0\}. \quad (5)$$

In order to provide a model taking into account the discretization in time, it is convenient to work with the sampling points that are the closest to the  $T_k$ 's, in other words to introduce:

$$P_0 \triangleq \{[T_n/\Delta t]; n = 1, \dots, M\},$$

and to relate (1) to a more standard regression model. More precisely, we can denote by  $\mathcal{R}$  the subset of non negative regressors as follows:

$$\mathcal{R} \triangleq \left\{ \beta \in \mathbb{R}_+^{Np} ; \forall j, \|\beta_j\|_1 = \sum_{\{n; [T_n]=t_j\}} E_n \right\}.$$

$\mathcal{R}$  is the set of candidate regressors related to the quantization of the energies  $E_n$  involved in (1). We thus define the optimal regressor  $\beta^0$  as the regressor in  $\mathcal{R}$  which fits optimally the actual noiseless model in terms of  $\ell_2$  norm, with regards to  $P_0$ :

$$\beta^0 \triangleq \underset{\beta \in \mathcal{R}, J(\beta)=P_0}{\text{Arg min}} \left\{ \sum_{i=0}^{N-1} \left( \sum_{n=1}^M E_n \Phi_n(t_i - T_n) - \mathbf{A} \beta_i \right)^2 \right\}. \quad (6)$$

We emphasize that the definition of the optimal regressor is by no means unique. However, the formulation in (6) is convenient for theoretical developments. Moreover, if  $\|\lambda\|_\infty \Delta t \ll 1$ , it is likely that  $\beta^0$  will be sparse, therefore relating to the method used in the rest of the paper.

Thus, the model investigated throughout the rest of the paper is

$$\mathbf{y} = \mathbf{A} \beta^0 + \varepsilon + \delta, \quad (7)$$

where  $\mathbf{y} \triangleq [y_0, y_1, \dots, y_{N-1}]^T$ ,  $\varepsilon \triangleq [\varepsilon_0, \varepsilon_1, \dots, \varepsilon_{N-1}]^T$ ,  $\beta^0$  is the sparse non negative vector whose non-null entries belong to the blocks indexed by  $P_0$ , as defined in (6),  $\delta \triangleq [\delta_0, \delta_1, \dots, \delta_{N-1}]^T$  is a vector quantifying the discretization error made between (1) and (7).

*Remark:* In this paper, we assumed that the variance of the additional noise,  $\sigma^2$ , is perfectly known. In practice, such assumption does not hold, but the noise variance of the spectrometer is usually estimated either during a preliminary calibration procedure, either when the detector is in idle state or between two consecutive pile-ups and records only noise. For the off-line processing of spectrometric data, or for very high counting rates, approaches performing the joint estimation of  $\beta^0$  and  $\sigma^2$  as appearing in [27], [28] should be preferred. These investigations are beyond the scope of the present paper, and should be discussed in future contributions.  $\square$

### B. Additional Notations and Assumptions

Alongside the rest of the paper, we shall use some notations from [25], which we recall and for convenience.

Given any finite discrete set  $I$ , we denote by  $|I|$  its cardinality. If  $I$  denotes a subset of  $\{0, 1, \dots, N-1\}$ , we denote by  $\mathbf{A}_I$  the submatrix of  $\mathbf{A}$  obtained by concatenation of time blocks whose index belong to  $I$ , namely

$$\mathbf{A}_I \triangleq [\dots, \mathbf{A}_k, \dots]_{k \in I}. \quad (8)$$

Given two subsets  $I, J$  of  $\{0, 1, \dots, N-1\}$ , we define the Gram matrix of size  $p|I| \times p|J|$  associated to  $\mathbf{A}_I$  and  $\mathbf{A}_J$  as

$$\mathbf{G}_{I,J} \triangleq \frac{1}{N} \mathbf{A}_I^T \mathbf{A}_J, \quad (9)$$

It is straightforward result that for the support length  $\tau$  introduced at (2), whenever  $|i-j| > \tau$  for all  $(i, j) \in I \times J$  then  $\mathbf{G}_{I,J} = 0$ . Two particular cases are to be further noted:

- whenever  $I = J$  and reduces to one element, one could drop the dependency in  $I, J$  in (9) and write simply  $\mathbf{G}$ , since by the very construction of  $\mathbf{A}$ , the Gram matrix of one timeblock is independent of the block index. We shall denote its minimal element by  $\mu \triangleq \min \mathbf{G}$ ; by construction,  $\mu > 0$ .
- whenever  $I = k; J = j$  are two distinct integers, then the maximal element of the matrix  $\mathbf{G}_{k,j}$  depends only upon the difference  $|k-j| = i$ , i.e. one could introduce the following function measuring the decreasing of the correlation between timeblocks:

$$\rho(i) \triangleq \max \mathbf{G}_{k,j} ; i = |k-j| \quad (10)$$

### III. NONPARAMETRIC LASSO-BASED ESTIMATOR OF THE ACTIVITY

Numerous estimators have been proposed in the literature for the estimation of the intensity of Poisson processes, both in homogeneous and non-homogeneous settings [29]–[31]. Nevertheless, all these methods assume a full knowledge of the sample path  $\{T_n, 1 \leq n \leq M\}$ , which is unavailable in our case. The suggested method can therefore be decomposed into two main steps: we first estimate the arrival times from  $\mathbf{y}$ , and the obtained values are afterwards plugged in into a standard kernel estimate of the intensity.

The method used for the estimation of arrival times (or, equivalently,  $P_0$ ) is the same as in [25], and is based on the non-negative LASSO [32], [33] (NNLASSO) which is introduced in the following optimization problem:

$$\hat{\beta}(r) = \underset{\beta \in \mathbb{R}_+^{Np}}{\text{Arg min}} \left\{ \frac{1}{2N} \|\mathbf{y} - \mathbf{A} \beta\|_2^2 + r \|\beta\|_1 \right\}, \quad (11)$$

where the tuning parameter  $r$  quantifies the trade-off between sparsity and estimation precision. The NNLASSO provides a sparse estimator  $\hat{\beta}(r)$  such that  $\mathbf{A} \hat{\beta}(r)$  approximates  $\mathbf{y}$ . We do not expect exact pattern recovery, since it relies upon restrictive conditions that are not met here. However, the NNLASSO pattern, being sparse for well-chosen values of  $r$ , should be concentrated nearby the support  $P_0$ .

In this paper, the intensity estimation procedure consists of plugging estimates of the arrival times in a nonparametric kernel estimate of the intensity, and will be denoted by NIL (Nonparametric Intensity using LASSO). The computation of  $\hat{\lambda}_{NIL}$  is detailed in Subsection III-B. Though this approach seems natural in light of [25], it should be emphasized that this plug-in approach raises numerous questions. Contrary to the homogeneous case, where only the greatest arrival times and the number of pulses were required to estimate the intensity, in the non-homogeneous setting we need to guarantee a good localization of the times estimates as well. This is particularly challenging, since it is well known that LASSO cannot provide consistency in selection unless irrepresentability conditions hold, which is not the case here. Furthermore, the sparsity parameter  $r$  and the parameters typically related to nonparametric kernel estimation (such as the type of kernel and its

bandwidth) should be carefully selected. In the following sections, we shall detail the estimation procedure and provide theoretical results and further insights on the proposed approach, such as the related estimation error.

### A. Results on the Estimation of the Arrival Times

We recall in Table I the notations used in the regression model and the estimation of the arrival times. These terms will appear in the following propositions.

| Variable name      | Definition                                                                                          | Assumptions/Remarks                                                                    |
|--------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| $E_n$              | Photonic energies                                                                                   | $e_{\min} \leq E_n \leq e_{\max}$ almost surely                                        |
| $\lambda(t)$       | Source activity                                                                                     | Bounded by $\ \lambda\ _\infty$                                                        |
| $\mathbf{A}$       | Dictionary                                                                                          | Made of translated shapes of durations $\tau \Delta t$                                 |
| $\sigma$           | Additive noise standard deviation                                                                   |                                                                                        |
| $\mathbf{A}_I$     | sub-dictionary of blocks $\mathbf{A}_j$ indexed by elements $i$ of $I \subset \{0, 1, \dots, N-1\}$ | Denoted by $\mathbf{A}_{i,j}$ if $I = \{i, j\}$                                        |
| $\beta^0$          | True regressor                                                                                      | Sparse vector                                                                          |
| $\delta$           | Discretization error between $\sum_{n=1}^M E_n \Phi(t_i - T_n)$ and $\mathbf{A}\beta_i^0$           |                                                                                        |
| $\hat{\beta}(r)$   | LASSO estimate of $\beta^0$                                                                         | LASSO estimate with a sparsity parameter $r$                                           |
| $J(\beta)$         | $\{0 \leq i \leq N-1; \beta_i \neq 0\}$                                                             | Block-sparsity pattern of any vector $\beta$                                           |
| $\mathbf{G}_{I,J}$ | Gram matrix $\frac{1}{N} \mathbf{A}_I^T \mathbf{A}_J$                                               | Denoted by $\mathbf{G}$ if $I = J$ , $\mathbf{G}_{i,j}$ if $I = \{i\}$ and $J = \{j\}$ |
| $\rho(i)$          | $\rho(i) = \max \mathbf{G}_{i,j}$                                                                   | Criterion of cross-correlation between timeblocks                                      |
| $\mu$              | $\mu = \min \mathbf{G}$                                                                             | Minimum cross-correlation between dictionary's shapes                                  |
| $\alpha$           | Thresholding level of the $\ \hat{\beta}_i(r)\ _1$                                                  |                                                                                        |

TABLE I

SUMMARY OF THE NOTATIONS RELATED TO THE REGRESSION MODEL AND THE ESTIMATIONS OF THE ARRIVAL TIMES

At each integer  $k$  corresponding to a pulse start one should observe a significantly high correlation between the signal and the columns of the  $k$ -th block  $\mathbf{A}_k$ ; more generally the same should hold for columns belonging to close surrounding blocks. This is why we expect that NNLASSO would select shapes inside several blocks close to the ones indexed by  $P_0$ . To begin with, we provide a bound on  $\max_{k \in P_0} \|\beta_k^0\|_1$ . It relies on a priory known bounds on the energy carried by a single photon.

**Lemma 1:** Let  $0 < \eta < 1$ . Define  $q$  as the least positive integer satisfying:

$$\frac{(\|\lambda\|_\infty \Delta t)^q}{q!} < \eta; \quad (12)$$

Conditionally to a given Poissonian sample, we have:

$$\mathbb{P} \left( \max_{k \in P_0} \|\beta_k^0\|_1 \leq q e_{\max} \right) \geq 1 - \eta \# P_0$$

*Proof:* See Appendix B. ■

The motivation of Lemma 1 can be explained briefly as follows: when the activity of the source is too high, the probability of having two or more photon arrival times between two sampling times increases. However, we know that NNLASSO performs poorly regarding consistency in selection. Therefore, Lemma 1 guarantees that at most  $q$  pulses start around the same sampling point, with a probability depending on the the actual sparsity parameter  $P_0$ .

Based on Lemma 1, Proposition 1 provides sufficient conditions so that some thresholded LASSO pattern is close to the subset  $P_0$  with a sufficiently high probability.

**Proposition 1:** Let  $0 < \eta < 1$ . Define  $q$  as in (12) from Lemma 1, and let  $m$  be a positive integer. If  $r$  and  $\alpha$  are chosen such that:

$$\alpha > 2\sigma\sqrt{\tau+1} + \frac{\|\delta\|_2}{\sqrt{N}}, \quad r > 2qe_{\max} \sum_{i>m} \rho(i),$$

then for any integer  $j$  satisfying the following threshold condition:

$$\|\hat{\beta}_j(r)\|_1 > \alpha, \quad (13)$$

there exists an element  $j_0$  in  $P_0$  such that  $|j - j_0| \leq m$ , with probability greater than  $1 - \eta \# P_0$ .

*Proof:* See Appendix C. ■

In the result given above,  $m$  denotes a level of confidence between the actual sparsity pattern  $P_0$  and the one derived from a thresholded version of the NNLASSO regressor  $\hat{\beta}(r)$ . Proposition 1 states that under a prior assumption on the activity, the choice of  $m$  induces a lower bound on the values of the sparsity parameter  $r$  and the threshold  $\alpha$ . With these settings, any sampling time  $t_j$  such that  $\hat{\beta}_j(r)$  satisfies (13) is close from at most  $m$  sampling points to some actual photon arrival time. Hence Proposition 1 indicates that a possible strategy to estimate  $P_0$  is to use such a thresholded version of  $\hat{\beta}(r)$ . Nevertheless, Proposition 1 can be considered somehow incomplete, since it does not guarantee that all the elements of  $P_0$  are close from at most  $m$  sampling points to a block  $\hat{\beta}_j(r)$  satisfying (13) as well. Furthermore, Proposition 1 raises another valid concern: the amplitudes of the electrical pulses depend on the photons' energies  $E_n$ , and if some pulse has a small amount of energy, it might simply go undetected due to the thresholding step. Eventually, this would result in an underestimation of the activity.

The following proposition answers to both concerns. It provides upper bounds on  $r$  and  $\alpha$  such that all the elements of  $P_0$  are close to a block  $\hat{\beta}_j(r)$  satisfying (13) with a high probability. Furthermore, it gives a condition on the minimal energy  $e_{\min}$  so that no pulse of small energy would be discarded by the thresholding operation.

**Proposition 2:** Assume that

$$\mu e_{\min} > 2\sigma\sqrt{\tau+1} + \frac{\|\delta\|_2}{\sqrt{N}}; \quad (14)$$

Suppose the sparsity parameter  $r$  is chosen satisfying:

$$0 < r < \mu e_{\min} - 2\sigma\sqrt{\tau+1} - \frac{\|\delta\|_2}{\sqrt{N}}; \quad (15)$$



with  $r$  set as before, let  $m$  be the smallest integer satisfying

$$\sum_{i>m} \rho(i) < \frac{\mu e_{\min} - 2\sigma\sqrt{\tau+1} - \frac{\|\delta\|_2}{\sqrt{N}} - r}{2 \max_j \|\hat{\beta}_j(r)\|_1}, \quad (16)$$

and  $\alpha$  satisfying

$$\frac{1}{2m+1} \left[ \mu e_{\min} - 2\sigma\sqrt{\tau+1} - \frac{\|\delta\|_2}{\sqrt{N}} - 2 \left( \sum_{i>m} \rho(i) \right) \max_j \|\hat{\beta}_j(r)\|_1 - r \right] > \alpha. \quad (17)$$

Then for all  $k$  in  $P_0$ , there exists at least one integer  $j$  such that  $\|\hat{\beta}_j(r)\|_1 \geq \alpha$  and  $|j-k| \leq m$ , with probability greater than  $1 - N \exp(-N(\tau+1)/2)$ .

*Proof:* See Appendix D. ■

In proposition 2, an assumption of the minimal energy of a pulse sets a condition for  $r$ ; for each such  $r$  one can associate a precision level  $m$ , and finally possible values for  $\alpha$ . Since we aim for pulse localization, the best possible precision is recommended, hence (16) could be seen as a way to select an optimal  $r$ . Indeed one should choose a value of  $r$  providing the highest right hand side in (16), because a quantity such as  $\sum_{i>m} \rho(i)$  decreases with  $m$ . Alternatively, we can remark that (14) can also be interpreted as a condition on the sampling period  $\Delta t$ , since  $\mu$  tends to increase as  $\Delta t$  decreases. Knowledge of the radioactive source (that is,  $e_{\min}$ ) defines which sampling frequency should be used to guarantee the localization of small electrical pulses. Conversely, if the sampling frequency of the apparatus is set, the practitioner should be aware that there is a minimum energy level below which some electrical pulses can go undetected.

We wish now to provide assumptions guaranteeing both conclusions of propositions 1 and 2. It is worth noting that in such a statement, the parameter choices should involve not just a prior lower bound  $e_{\min}$ , but also a prior upper bound  $e_{\max}$ , on the individual energies. This can be easily understood: the higher the energy is, the higher the points of the associated electrical pulse will be, even those remote from its starting point. This is likely to make NN-LASSO put coefficients inside single electrical pulses; this situation impinges seriously upon time localization.

*Proposition 3:* Assume the following:

$$\mu e_{\min} > 2(\tau+1) \left[ 2\sigma\sqrt{\tau+1} + \frac{\|\delta\|_2}{\sqrt{N}} \right] \quad (18)$$

Let  $r$  be a positive parameter such that:

$$r < \mu e_{\min} - 2(\tau+1) \left[ 2\sigma\sqrt{\tau+1} + \frac{\|\delta\|_2}{\sqrt{N}} \right] \quad (19)$$

To such a choice of  $r$  associate the smallest integer  $m \leq \tau$  such that both inequalities hold:

$$\sum_{i>m} \rho(i) < \frac{r}{2qe_{\max}} \quad (20)$$

$$\sum_{i>m} \rho(i) < \frac{\mu e_{\min} - 2(m+1) \left[ 2\sigma\sqrt{\tau+1} - \frac{\|\delta\|_2}{\sqrt{N}} \right] - r}{2 \max_j \|\hat{\beta}_j(r)\|_1}; \quad (21)$$

Set now the threshold value  $\alpha$  so that

$$\begin{aligned} & \frac{1}{2m+1} \left[ \mu e_{\min} - 2\sigma\sqrt{\tau+1} - \frac{\|\delta\|_2}{\sqrt{N}} - 2 \left( \sum_{i>m} \rho(i) \right) \max_j \|\hat{\beta}_j(r)\|_1 \right] \\ & > \alpha > 2\sigma\sqrt{\tau+1} + \frac{\|\delta\|_2}{\sqrt{N}} \end{aligned} \quad (22)$$

then both conclusions of Proposition 2 and Proposition 1 hold under probability  $1 - \eta \#P_0 - N \exp(-N(\tau+1)/2)$ .

*Proof:* The first condition (18) about the minimal energy concentrated at an integer of  $P_0$  makes possible the indicated choices of  $r$  and  $\alpha$ . The rest follows from the other propositions above. ■

## B. Overview of the Method for NHPP Intensity Estimation

In this section we assume that conditions of proposition 3 are met, and that accordingly we set the parameter  $r$ , the threshold value  $\alpha$ , and the integer  $m$  used as a confidence neighbourhood size for pulse localization. From the previous results, two distinct integers  $j_1, j_2$  in  $J(\hat{\beta}(r))$  such that  $\|\hat{\beta}_{j_1}(r)\|_1 > \alpha$  and  $\|\hat{\beta}_{j_2}(r)\|_1 > \alpha$  cannot be related to one single element of  $P_0$  as long as they are distant from at least  $2m+1$ . Therefore we shall consider a maximal length sequence of elements in  $J(\hat{\beta}(r))$ , which are pairwise distant from  $2m+1$  to the least:

$$\begin{aligned} \widehat{M} &\triangleq \max \left\{ M \in \mathbb{N} ; \exists j_1 < \dots < j_M \in J(\hat{\beta}(r)) \right. \\ &\quad \left. \text{s.t. } \|\hat{\beta}_{j_k}(r)\|_1 > \alpha \text{ and } j_{k+1} - j_k > 2m \ \forall k \leq M \right\}; \end{aligned} \quad (23)$$

denote also by  $\{\hat{j}_i, 1 \leq i \leq \widehat{M}\}$  a sequence of elements in  $J(\hat{\beta}(r))$  satisfying both thresholding and distance criteria and reaching the length that is specified in (23). Our estimated arrival times shall be defined by  $\hat{T}_i = \hat{j}_i \Delta t$ ,  $i = 1, \dots, \widehat{M}$ . To further estimate the activity  $\lambda(t)$ , we plug the estimated arrival times  $\{\hat{T}_i, i = 1, \dots, \widehat{M}\}$  into the Lewis and Shedler [30] nonparametric estimator as follows:

$$\hat{\lambda}_{NIL}(t) = \frac{1}{h} \sum_{i=1}^{\widehat{M}} W \left( \frac{t - \hat{T}_i}{h} \right), \quad (24)$$

where  $W$  is a kernel function and  $h > 0$  is a user-defined bandwidth parameter. The following assumptions on the kernel function  $W$  are made:

- 1)  $W$  is defined on a symmetric compact support  $[-\ell(W)/2; \ell(W)/2]$  for some positive number  $\ell(W)$ , such that  $\int_{-\infty}^{\infty} W(t) dt = 1$ .
- 2)  $W$  has a bounded derivative.

These assumptions shall be used to investigate the behaviour of the error between  $\lambda$  and  $\hat{\lambda}_{NIL}$ .

### C. Error Decomposition

In this section we aim to bound the final error term  $\|\hat{\lambda}_{NIL} - \lambda\|_2$  by decomposing it as a sum of terms we can control. It is well known that when  $t$  is getting too close to the boundaries of the observational time window  $[0, T]$ , a bias appears in the estimation due to the lack of data to estimate  $\lambda(t)$  accurately [34]. We shall therefore make use for the error of the  $\ell_2$  distance between two square integrable functions  $f, g$  both defined on the restricted interval  $[h\ell(W)/2; T - h\ell(W)/2]$

i.e.:  $\|f - g\|_2 = \left( \int_{h\ell(W)/2}^{T-h\ell(W)/2} (f(t) - g(t))^2 dt \right)^{1/2}$ . We shall

first introduce the intermediate estimator  $\hat{\lambda}_{oracle}$ : in practice, the only information we could get from the sampled signal  $y$  resides in the rounded beginnings of individual pulses, so the oracle estimator can be defined as follows

$$\hat{\lambda}_{oracle}(t) = \frac{1}{h} \sum_{k \in P_0} W\left(\frac{t - t_k}{h}\right). \quad (25)$$

This estimator relies on the sole random pattern  $P_0$ , and does not take into account (possible) multiple pulses relating to one single sampling time. Hence, as a first step one could bound the error with the following two terms:

$$\|\lambda - \hat{\lambda}_{NIL}\|_2 \leq \|\lambda - \hat{\lambda}_{oracle}\|_2 + \|\hat{\lambda}_{oracle} - \hat{\lambda}_{NIL}\|_2 \quad (26)$$

Next proposition bounds the expectation of the first error term  $\|\lambda - \hat{\lambda}_{oracle}\|_2$ , which is so far unrelated to any particular method of sample estimation. A bound on the term  $\|\hat{\lambda}_{oracle} - \hat{\lambda}_{NIL}\|_2$  is the matter of proposition 5 and corollary 1.

**Proposition 4:** Assume that  $\|\lambda\|_{\infty} \Delta t < 3$ . Then:

$$\begin{aligned} & \mathbb{E}[\|\lambda - \hat{\lambda}_{oracle}\|_2] \\ & \leq \frac{\|W\|_2^2}{h} \int_0^T \lambda(t) dt + \|\lambda\|_{\infty}^2 (T - \ell(W)h) \\ & \quad + \frac{\Delta t \sqrt{\Delta t + h\ell(W)}}{2h^2} \|W'\|_{\infty} \int_0^T \lambda(t) dt \\ & \quad + \frac{\|W\|_2}{2\sqrt{h}} \|\lambda\|_{\infty} \Delta t \int_0^T \lambda(t) dt \end{aligned} \quad (27)$$

*Proof:* See Appendix E.

We now provide in the next proposition an error bound for  $\|\hat{\lambda}_{NIL} - \hat{\lambda}_{oracle}\|_2$ .

**Proposition 5:** Under the assumptions of Proposition 2 the corresponding estimator  $\hat{\lambda}_{NIL}$  satisfies the following:

$$\begin{aligned} & \|\hat{\lambda}_{NIL} - \hat{\lambda}_{oracle}\|_2 \\ & \leq \frac{\|W\|_2}{\sqrt{h}} (\#P_0 - \widehat{M}) \\ & \quad + \widehat{M} \frac{\|W'\|_{\infty} \Delta t}{h^2} m \sqrt{2m\Delta t + h\ell(W)} \end{aligned} \quad (28)$$

*Proof:* Given any  $i = 1, \dots, \widehat{M}$  let us associate the time  $t_{k(i)}$ , for  $k(i) \in P_0$ , which is the closest to  $\widehat{T}_i$ : obviously this defines a one-to-one correspondence between the set  $\{\widehat{T}_i; i = 1, \dots, \widehat{M}\}$  and some subset of  $P_0$  we shall denote by  $\widehat{P}_0$ . By proposition 1 the following bound holds for each difference:  $\left| W\left(\frac{t - \widehat{T}_i}{h}\right) - W\left(\frac{t - t_{k(i)}}{h}\right) \right| \leq \|W'\|_{\infty} \frac{m\Delta t}{h}$ , for all  $t$  and  $i = 1, \dots, \widehat{M}$ , and such a difference is nonzero only when  $|t - \widehat{T}_i| < h \frac{\ell(W)}{2} + m\Delta t$ ; so that we have:

$$\begin{aligned} & \left\| W\left(\frac{t - \widehat{T}_i}{h}\right) - W\left(\frac{t - t_{k(i)}}{h}\right) \right\|_2 \\ & \leq \|W'\|_{\infty} \frac{m\Delta t \sqrt{2m\Delta t + h\ell(W)}}{h}; \end{aligned}$$

this immediately yields the following:

$$\begin{aligned} & \|\hat{\lambda}_{NIL} - \hat{\lambda}_{oracle}\|_2 \\ & \leq \frac{1}{h} \left\| \sum_{i=1}^{\widehat{M}} W\left(\frac{t - \widehat{T}_i}{h}\right) - \sum_{k \in P_0} W\left(\frac{t - t_k}{h}\right) \right\|_2 \\ & \leq \frac{1}{h} \sum_{i=1}^{\widehat{M}} \left\| W\left(\frac{t - \widehat{T}_i}{h}\right) - W\left(\frac{t - t_{k(i)}}{h}\right) \right\|_2 \\ & \quad + \frac{1}{h} \sum_{k \in P_0 \setminus \widehat{P}_0} \left\| W\left(\frac{t - t_k}{h}\right) \right\|_2 \\ & \leq \widehat{M} \|W'\|_{\infty} \frac{m\Delta t \sqrt{2m\Delta t + h\ell(W)}}{h^2} \\ & \quad + \frac{1}{\sqrt{h}} (\#P_0 - \widehat{M}) \|W\|_2 \end{aligned}$$

As a remark, note that though  $\widehat{M}$  can be computed from the NNlasso regressor  $\widehat{\beta}(r)$ , it seems difficult to bound it optimally, especially from below. Besides, if conditions of Proposition 3 are met, then  $\widehat{M}$  is lower bounded by the length of a maximal subsequence of Poissonian times which are pairwise distant from  $4m + 1$  to the least when rounded, i.e.

$$\begin{aligned} & \widehat{M} \geq \max \left\{ K ; \exists i_1 < i_2 < \dots < i_K \text{ all in } P_0 \right. \\ & \quad \left. \text{s.t. } i_{j+1} - i_j > 4m \ \forall j \right\} \end{aligned}$$

■ We shall denote by  $P_0^{(m)}$  any sub-sequence of  $P_0$  whose length reaches the maximum indicated above. The next corollary follows immediately from proposition 5.

*Corollary 1:* Suppose all conditions of proposition 3 are satisfied, and assume the kernel function  $W$  and the bandwidth  $h$  satisfy

$$\left( \frac{\|W'\|_\infty \Delta t}{\|W\|_2} m \right)^2 < \frac{h^3}{2m\Delta t + h\ell(W)} ;$$

then the following holds:

$$\begin{aligned} & \|\hat{\lambda}_{NIL} - \hat{\lambda}_{oracle}\|_2 \\ & \leq \frac{\|W\|_2}{\sqrt{h}} (\#P_0 - \#P_0^{(m)}) \\ & \quad + \#P_0^{(m)} \frac{\|W'\|_\infty \Delta t}{h^2} m \sqrt{2m\Delta t + h\ell(W)} \end{aligned} \quad (29)$$

*Remark:* The result can be related to the result discussed in [27], that good estimation (in the sense of adaptivity) can be performed if  $\#J(\beta^0) \log(Np/\#J(\beta^0)) < N$ . Indeed, this inequality can hold provided  $\|\lambda(t)\|_\infty$  is sufficiently small. Moreover, it can be observed that all along the proofs of the different results, we made no use of the fact that  $\Delta t$  is constant. This assumption can indeed be relaxed by assuming only that  $\Delta t$  is bounded with probability one. This guarantees that results of similar nature remain valid, even in cases of jitter from the Analog-Digital converter.  $\square$

#### IV. APPLICATIONS

In this section, we present results obtained from simulations. The overall implementation was done in Matlab, and the non-negative LASSO implementation is taken from the SparseLab library, available at <http://sparselab.stanford.edu/>.

##### A. Results on Simulations

The intensity of the Poisson process is chosen equal to

$$\lambda(t) \triangleq 0.2e^{-(t-30)^2/75} + xe^{-(t-60)^2/50}$$

on  $[0, 100]$ , where  $x$  is a tuning parameter chosen between 0.1 and 3 in order to investigate the performances of the approach for high to very high counting rates. The sampling rate of the signal is taken equal to  $\Delta t = 1$ . The energies  $\{E_n, n \geq 0\}$  are drawn according to a truncated Gaussian density with mean 700 and standard deviation 10, with support  $[670, 730]$ . The noise variance is chosen such that  $\sigma = 1$ , in order to reflect the excellent Signal Noise Ratios (SNRs) encountered in the case of HPGe detectors commonly used in the field. The electrical pulses' shapes were created accordingly to (2), and two cases were investigated: in the first one, shapes were drawn randomly from  $\mathbf{A}$ ; in the second case, Gamma shapes were created with parameters  $\theta_1, \theta_2$  chosen randomly accordingly to a uniform distribution on  $[0.3, 1.3]$ . 20 different shapes (and their shifted versions) were used in the dictionary  $\mathbf{A}$ , with parameters  $\theta_1 = \theta_2$  ranging from 0.4 to 1.2. This choice illustrates the fact that  $\mathbf{A}$  must be carefully chosen, though some discrepancy between the shapes appearing in  $\mathbf{y}$  and  $\mathbf{A}$  may be allowed. We chose a truncated Gaussian kernel for  $W$ . A data-driven choice of the bandwidth  $h$  remains an important problem in practice [35]; since this question is out of the scope of the present paper, we just set  $h = 2$ . For both

cases, we set  $\eta = 0.05$ ,  $m = 2$ ,  $r = 3qe_{max} \sum_{i \geq 2} \rho(i)$  and  $\alpha = 3\sigma\sqrt{\tau+1}$ . This guarantees that Propositions 1. and 2. hold, provided  $\|\delta\|_2 \leq \sqrt{N}\sigma\sqrt{\tau+1}$ .

We compared, in the simulations, the asymptotic performances of the following activity estimates: our LASSO-based estimate  $\hat{\lambda}_{NIL}$ ;  $\hat{\lambda}_{ideal}$  described in (32) (denoted by *ideal estimate*);  $\hat{\lambda}_{oracle}$  described in (25) (denoted by *oracle estimate* in the results), and the estimator of [30], when the arrival times are detected by a simple threshold (denoted by *uncorrected estimate*). In each case, 10000 simulations were performed. We present in our figures the average of the activity estimates in each case, and an approximation of the Mean Integrated Squared Error (MISE) between the actual activity and its estimate.

Results in the case of signals drawn randomly from the dictionary are displayed in Figure 2; the average estimates are shown in Figure 2a, and the approximate MISE, in Figure 2b. We can observe that the proposed method provides an estimate which behaves both like (25) with a slight bias. Results in the case of signals with random parameters not taken from the dictionary are displayed in Figure 3; the average estimates are shown in Figure 3a, and the approximate MISE, in Figure 3b.

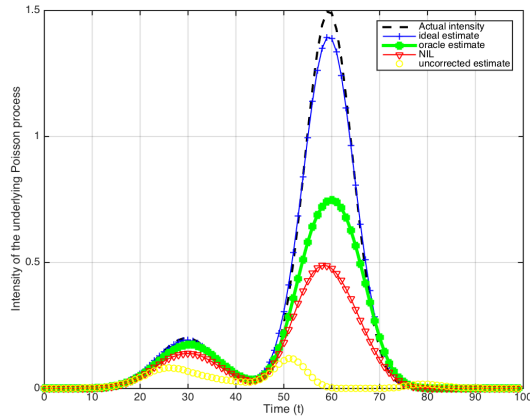
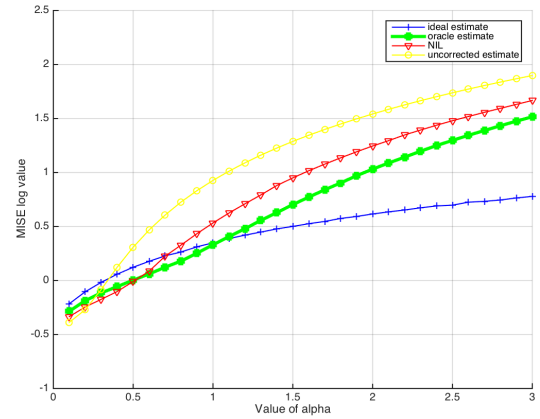
In order to empirically evaluate the efficiency of the proposed method, we show in Figures 4 and 5 the 95% confidence bands computed from our Monte-Carlo experiment for the LASSO-based, oracle and ideal estimates.

##### B. Influence of the dictionary-signal mismatch

In order to investigate the perturbations stemming from the inadequacy between the dictionary  $\mathbf{A}$  and the signal  $\mathbf{y}$ , the following experiment is performed: the dictionary is generated as in the random setting, with the shapes' parameters being drawn accordingly to a uniform distribution on  $[0.4; 1.3]$  (again, we choose  $\theta_1 = \theta_2$ ), the number of shapes being  $p = 10$ , and  $\alpha = 1.5$ . On the other hand, the pulses' shapes of  $\mathbf{y}$  are simulated accordingly to Gamma shapes with  $\theta_1 = \theta_2$ , where  $\theta_1$  is drawn accordingly to a uniform distribution on  $[a, b]$ . We investigated two different cases separately: in the first setting, the shapes of  $\mathbf{y}$  have a smaller support than  $\mathbf{A}$ 's, that is we tried to recover electrical pulses with a very steep decrease from a dictionary including smoother shapes; on the contrary, in the second setting the shapes of  $\mathbf{y}$  had a larger support than  $\mathbf{A}$ 's. The mismatch between the shapes from  $\mathbf{A}$  and the ones appearing in  $\mathbf{y}$  was characterized by the Jaccard index between  $[a, b]$  and  $[0.4, 1.3]$ , that is

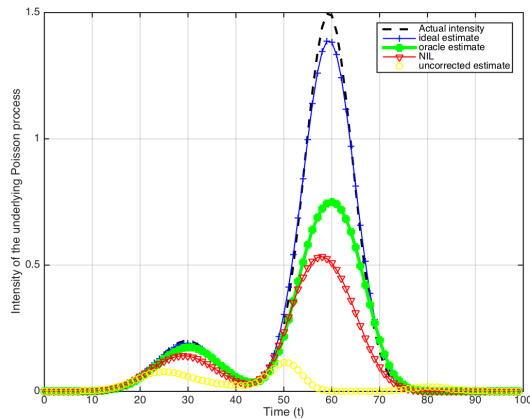
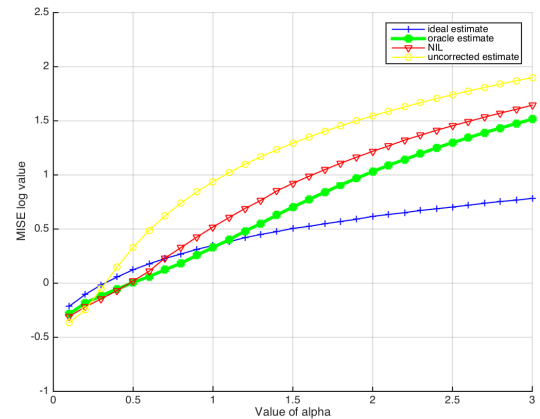
$$\begin{cases} 0 & \text{if } [a, b] \cap [0.4, 1.3] = \emptyset, \\ \frac{\min\{b, 1.3\} - \max\{0.4, a\}}{\max\{b, 1.3\} - \min\{a, 0.4\}} & \text{otherwise.} \end{cases}$$

For example, a Jaccard index of 0.6 in the first setting can be obtained with  $a = 0.175$ ,  $b = 1.075$ , while in the second setting it can be obtained with  $a = 0.625$ ,  $b = 1.525$ . A Jaccard index of 0 means that the parameters  $\theta_1, \theta_2$  of the shapes in  $\mathbf{A}$  have no connection whatsoever with  $\mathbf{y}$ , whereas 1 means that  $a = 0.4$ ,  $b = 1.3$ . Observe that it does not mean that the shapes of  $\mathbf{A}$  perfectly match  $\mathbf{y}$ , but rather that the similarity between

(a) Average estimates for  $x = 1.5$ .

(b) Approximated MISE (log values).

Fig. 2. Results with shapes taken from the dictionary

(a) Average estimates for  $x = 1.5$ .

(b) Approximated MISE (log values).

Fig. 3. Results with Gamma shapes with randomly drawn parameters

shapes of  $\mathbf{A}$  and shapes of  $\mathbf{y}$  is the most likely to happen. The physical interpretation of a Jaccard index of 1 is that we have perfect knowledge of the detector, and know exactly what is the set of possible parameters of the shapes appearing in  $\mathbf{y}$ , while not being able to infer the exact values. Experimental results are presented in Table II. Not surprisingly, the best results in terms of MISE are obtained as long as there is adequacy between  $\mathbf{A}$  and  $\mathbf{y}$ , even partial, but both cases are asymmetrical. In the first setting, we tried to reconstruct sharp, localized pulses with much smoother shapes; LASSO thus provides a less sparse, spread solution, which cannot be corrected by the post-processing steps. On the other hand, in the second situation we estimate pulses smoother than the dictionary's shapes; though the solution provided by LASSO is less sparse for lower Jaccard indexes than for full adequacy, the non-null coefficients of  $\hat{\beta}(r)$  remain localized around the actual arrival times, and the post-processing stage deals with most of the perturbations. These considerations are illustrated in Figure 6. Therefore, though we observed a degradation of

| First setting |         | Second setting |        |
|---------------|---------|----------------|--------|
| Jaccard Index | MISE    | Jaccard Index  | MISE   |
| 0.00          | 25.9027 | 0.00           | 6.3145 |
| 0.10          | 8.3108  | 0.10           | 6.1290 |
| 0.20          | 6.7695  | 0.20           | 6.1510 |
| 0.30          | 6.6925  | 0.30           | 6.1194 |
| 0.40          | 6.7044  | 0.40           | 6.0617 |
| 0.50          | 6.5703  | 0.50           | 6.0934 |
| 0.60          | 6.5717  | 0.60           | 6.0592 |
| 0.70          | 6.4808  | 0.70           | 6.0105 |
| 0.80          | 6.4081  | 0.80           | 6.0463 |
| 0.90          | 6.2873  | 0.90           | 5.9181 |
| 1             | 5.9013  | 1              | 5.9013 |

TABLE II  
INFLUENCE OF DICTIONARY'S INADEQUACY ON THE MISE

the MISE, it was less in the second setting than in the first one. The choice of  $p$ , though important, was less critical in that matter. We observed, in numerical simulations, that provided



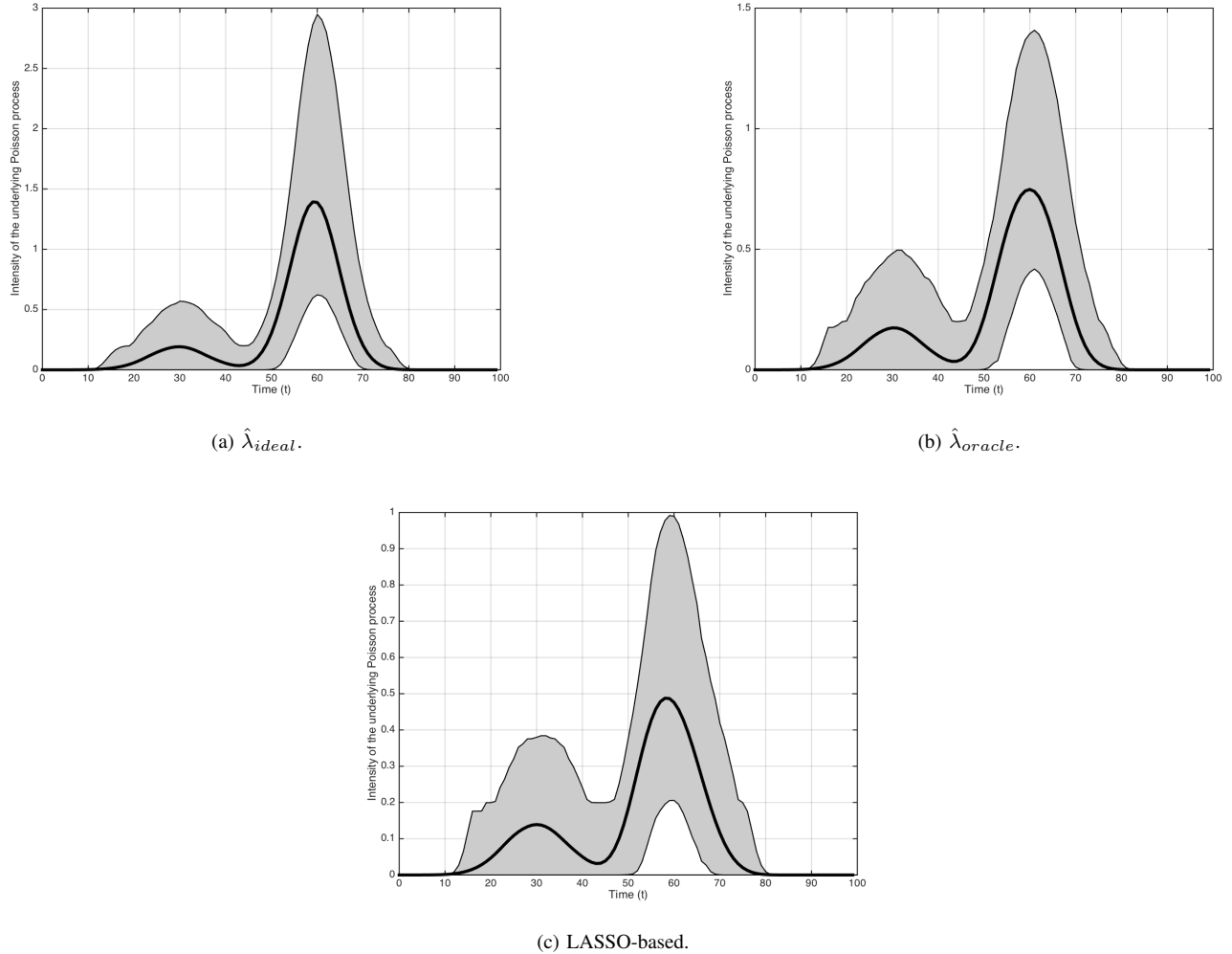


Fig. 4. 95% confidence intervals of the compared estimators with shapes taken from **A**

the shapes are drawn similarly for **A** and **y** (that is, with a Jaccard index of 1), there is little variation in the MISE when the number of shapes  $p$  goes above 20. This mostly is due to the fact that LASSO is not consistent in selection, and that  $\theta_1, \theta_2 \sim \mathcal{U}([0.4, 1.3])$  yield very similar shapes. However, in the cases of HPGc detectors used in the field, there is very little variation from one electrical pulse to the other, making this choice of parameters valid in practice.

### C. Results Using Real Data

For real data experiments, it is of interest to provide well-localized arrival times. In this paper, we present a result obtained on the signal recorded during the Pidie experiment, realized in CEA Saclay, France, in which a mixture of Americium and Cesium with high activity was used for the measurements. In that framework, a single minute of signal may include several millions of electrical pulses recorded. It is then of importance to obtain well-localized arrival times. Figures 7a and 7b illustrate the obtained results regarding the estimation of the arrival times.

### D. Discussion

It can be observed that  $\hat{\lambda}_{NIL}(t)$  performs relatively well when the shapes are drawn from the dictionary, and slightly less when the shapes' parameters are drawn randomly. Nonetheless, it can be seen in both cases that it performs almost as well as  $\hat{\lambda}_{oracle}$ , the latter being the best estimate of  $\lambda(t)$  that we can hope for after the signal's digitization. For higher values of  $x$ , we observe a loss in performance, and the MISE increases. This, however, is not surprising: for high values of  $\lambda(t)$ , the probability of obtaining several arrivals between two sampling points increases, thus leading to an underestimation of the actual intensity. We emphasize that, even in that case, the LASSO-based and the oracle perform quite similarly. Therefore, the relatively poor results obtained in that case are inherent to the sampling, and can in practice be improved only by better instrumentation. It is nevertheless noticed, both from the MISE investigation and from the computation of the confidence intervals, that the proposed method compares well with oracle procedures, thus making it a valid candidate for real spectrometric applications.

The proposed approach tends to overestimate the number

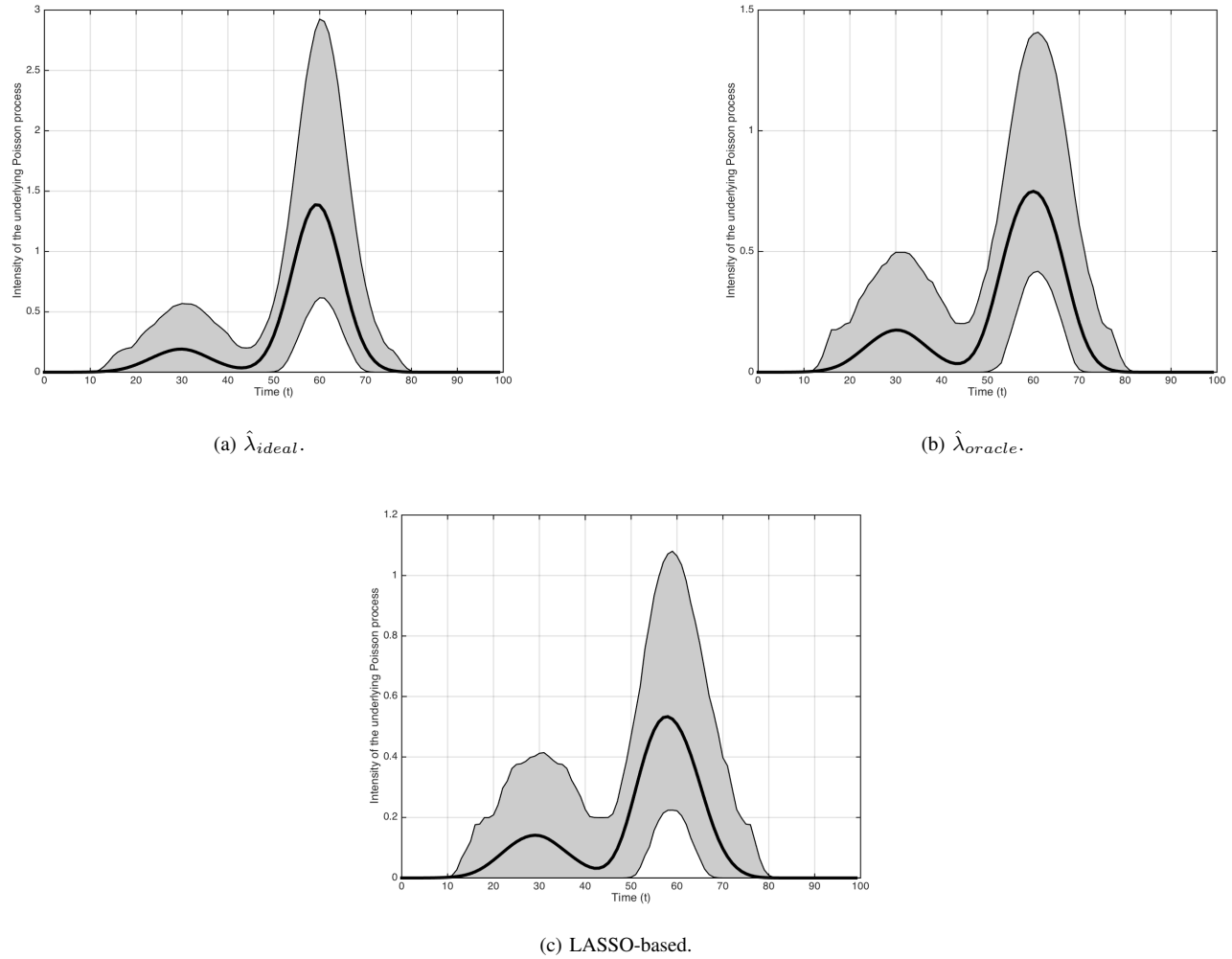
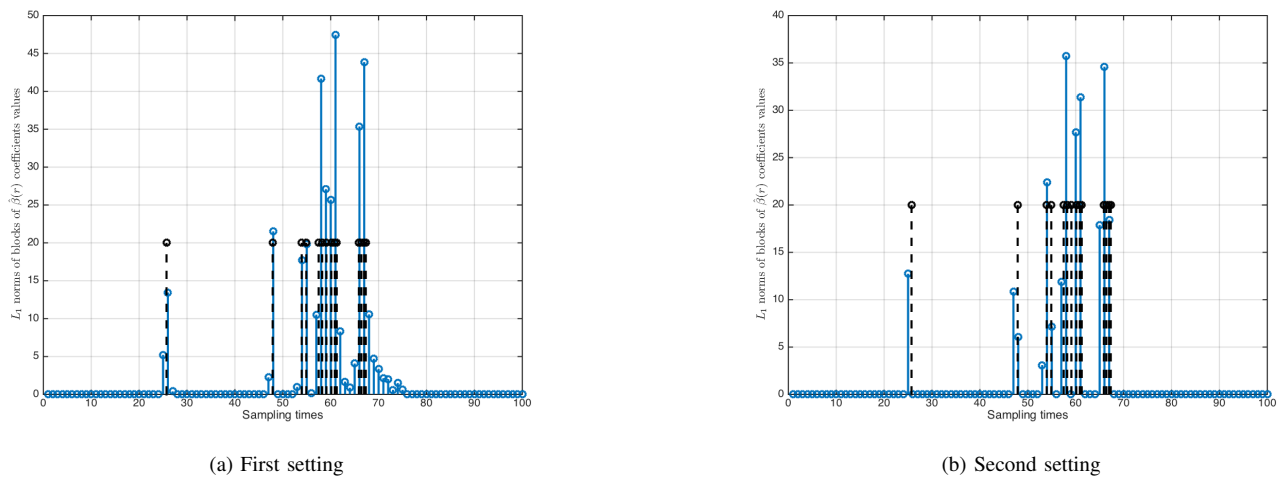
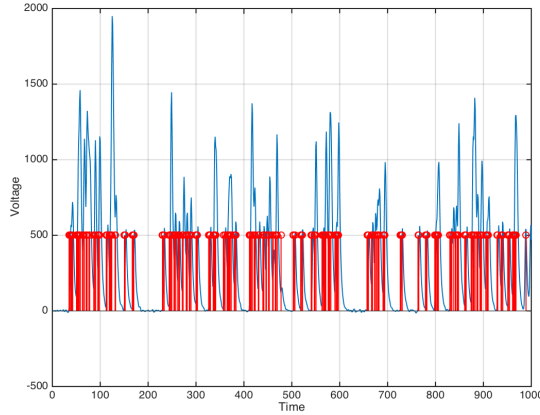


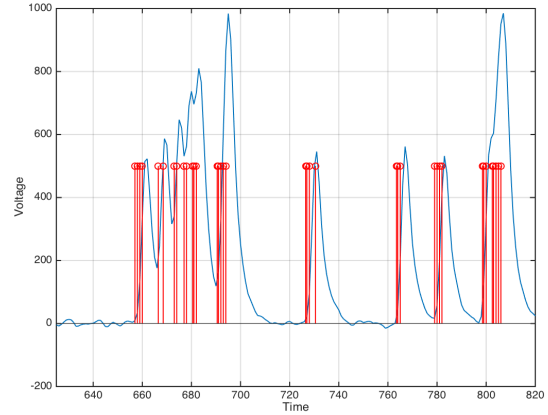
Fig. 5. 95% confidence intervals of the compared estimators with random shapes

Fig. 6. Comparison between the two settings, for a common Jaccard index of 0.1. The  $L_1$  norm of the blocks of  $\hat{\beta}(r)$  (solid blue) is compared to the actual arrival times (dotted black).

of arrival times, as noticed from the simulations and the results obtained on real data. In practice, an underestimated error term  $\|\delta\|_2$  leads the user to plug a weaker value of  $\alpha$  in Proposition 1; this yields a potential overestimation of



(a) signal (blue) and arrival times (red).



(b) Zoom on the 625-820 area.

Fig. 7. Results of the LASSO-based arrival times estimation on the Pidie dataset.

$\lambda$ . In addition, the choice of the sparsity parameter  $r$  in the LASSO regression problem (11) also bears consequences: by choosing too high a value of  $r$ , the estimation procedure is build upon a regressor  $\hat{\beta}(r)$  with poor  $\ell_1$  norm, adding to potential overestimation. However, the selected times are all well-localized around the true arrival times – accordingly to the main Propositions – and well-separated, even in case of very close pulses as shown in Figure 7b. Though of undeniable interest, a more refined selection of the arrival times is not in the scope of the present paper, and will be investigated in future contributions.

## V. CONCLUSION

In this paper, we presented a novel algorithm for the estimation of the intensity of a nonhomogeneous Poisson process, and illustrated the efficiency of the proposed approach for the case of gamma spectroscopy application. Practical conditions were derived, which guarantee the good behavior of the computed estimator. However, since this paper's emphasis is more on practicality rather than on optimality, the bounds derived should be refined in order to obtain optimal error bounds. This aspect will be investigated in future contributions.

## ACKNOWLEDGMENT

We would like to thank the reviewers for their thorough reading and remarks, which greatly improved the readability of the present paper.

## APPENDIX

### A. Additional Technical Lemmas

The following lemmas will be used alongside the paper.

**Lemma 2 (Mallat [36]):** For any white noise vector  $\varepsilon$  of length  $M$  and of unit variance; it holds that

$$\mathbb{P}(\|\varepsilon\|_2 \geq \sqrt{M} + t) \leq \exp(-t^2/2), \forall t > 0$$

**Lemma 3:** Let us write the model introduced in (7) as  $\mathbf{y} = \mathbf{A}\beta + \mathbf{w}$  with  $\mathbf{w} \triangleq \delta + \varepsilon$  as the overall perturbation term.

Given  $m < \tau \leq N$  an integer, it holds for all subsets  $J$  of  $m$  consecutive integers that:

$$\left\| \frac{\mathbf{A}_J^T \mathbf{w}}{N} \right\|_1 \leq mp \left[ 2\sigma\sqrt{\tau+1} + \frac{\|\delta\|_2}{\sqrt{N}} \right] \quad (30)$$

with probability greater than  $1 - N \exp(-N(\tau+1)/2)$ .

*Proof:* Given  $J$  as in the proposition, and  $j \in J$ , we get

$$\begin{aligned} \left\| \frac{\mathbf{A}_J^T \mathbf{w}}{\sqrt{N}} \frac{\mathbf{w}}{\sqrt{N}} \right\|_1 &\leq mp \frac{\|\mathbf{w}[j; \dots; j+\tau]\|_2}{\sqrt{N}} \\ &\leq mp \left[ \frac{\|\delta\|_2}{\sqrt{N}} + \frac{\|\varepsilon[j; \dots; j+\tau]\|_2}{\sqrt{N}} \right] \end{aligned}$$

We now apply Lemma 2 to  $\varepsilon[j; \dots; j+\tau]/\sigma$ , it follows that for any  $t > 0$

$$\begin{aligned} \mathbb{P} \left( \frac{\|\varepsilon[j; \dots; j+\tau]\|_2}{\sqrt{N}} \leq \frac{\sigma(\sqrt{\tau+1} + t)}{\sqrt{N}} \right) \\ \geq 1 - \exp(-t^2/2) \end{aligned} \quad (31)$$

Taking  $t = \sqrt{(\tau+1)N}$  in (31) yields

$$\begin{aligned} \mathbb{P} \left( \frac{\|\varepsilon[j; \dots; j+\tau]\|_2}{\sqrt{N}} \leq 2\sigma\sqrt{\tau+1} \right) \\ \geq 1 - \exp(-N(\tau+1)/2), \end{aligned}$$

and the lemma follows from the fact there are  $N - m$  subsets  $J$  of  $m$  consecutive integers. ■

### B. Proof of Lemma 1

In this proof we shall introduce for all integer  $k$  the following local Poissonian parameter

$$\Lambda_k \triangleq \int_{(k-1/2)\Delta t}^{(k+1/2)\Delta t} \lambda(t) dt$$

From the following easy inequalities:

$$\begin{aligned} & \exp(-\Lambda_k) \sum_{s>q} \frac{(\Lambda_k)^s}{s!} \\ & \leq \exp(-\Lambda_k) \frac{(\Lambda_k)^q}{q!} \left[ \Lambda_k + \frac{\Lambda_k^2}{2!} + \dots \right] \\ & \leq \frac{(\Lambda_k)^q}{q!} [1 - \exp(-\Lambda_k)] \end{aligned}$$

combined with the obvious inequality  $\Lambda_k \leq \|\lambda\|_\infty \Delta t$  one deduces the following conditional probability

$$\mathbb{P}(\#\{n; [T_n] = t_k\} > q \mid k \in P_0) \leq \frac{(\Lambda_k)^q}{q!} \leq \frac{(\|\lambda\|_\infty \Delta t)^q}{q!}$$

meaning that conditionally to  $k \in P_0$  there are more than  $q$  pulses starting around  $t_k$  under probability at most  $\eta$  by assumption (12). The conclusion follows immediately.

### C. Proof of Proposition 1

Let  $j \in J(\hat{\beta}(r))$  such that  $\|\hat{\beta}_j(r)\|_1 > \alpha$ : thus for some index  $1 \leq s \leq p$  it holds by the Karush–Kuhn–Tucker (KKT) conditions and by Lemma 3 (setting there  $m = p = 1$ ) that under probability greater than  $1 - N \exp(-N(\tau + 1)/2)$ :

$$\begin{aligned} \alpha + r & < \frac{1}{N} \mathbf{A}_{j;s}^T \mathbf{A} \hat{\beta}(r) + r \\ & = \frac{1}{N} \mathbf{A}_{j;s}^T \mathbf{A} \beta^0 + \frac{1}{N} \mathbf{A}_{j;s}^T (\varepsilon + \delta) \\ & \leq \frac{1}{N} \mathbf{A}_{j;s}^T \mathbf{A} \beta^0 + 2\sigma\sqrt{\tau+1} + \frac{\|\delta\|_2}{\sqrt{N}} \end{aligned}$$

If there is no integer  $k \in P_0$  such that  $|k-j| \leq m$  the previous inequality implies

$$\begin{aligned} \alpha + r - 2\sigma\sqrt{\tau+1} - \frac{\|\delta\|_2}{\sqrt{N}} & < \sum_{k: |k-j| > m} \rho(|k-j|) \|\beta_k^0\|_1 \\ & \leq \left( \max_{k \in P_0} \|\beta_k^0\|_1 \right) 2 \sum_{i>m} \rho(i) \end{aligned}$$

In other words, if we have:

$$2\sigma\sqrt{\tau+1} + \frac{\|\delta\|_2}{\sqrt{N}} + \left( \max_{k \in P_0} \|\beta_k^0\|_1 \right) 2 \sum_{i>m} \rho(i) < \alpha + r$$

then the distance from  $j$  to the pattern  $P_0$  is  $m$  to the most. The rest follows from Lemma 1.

### D. Proof of Proposition 2

By the KKT conditions applied to the regressor  $\hat{\beta}(r)$  and the triangular inequality, it holds that for any  $k$ :

$$\begin{aligned} & \sum_{|j-k| \leq m} \|\hat{\beta}_j(r)\|_1 + \frac{1}{pN} \left\| \mathbf{A}_k^T \sum_{\{j: |j-k| > m\}} \mathbf{A}_j \hat{\beta}_j(r) \right\|_1 \\ & \geq \frac{1}{pN} \left\| \mathbf{A}_k^T \mathbf{A} \hat{\beta}(r) \right\|_1 \geq \frac{1}{pN} \left\| \mathbf{A}_k^T \mathbf{y} \right\|_1 - r \end{aligned}$$

which implies the following

$$\begin{aligned} & \sum_{|j-k| \leq m} \|\hat{\beta}_j(r)\|_1 + 2 \left[ \sum_{i>m} \rho(i) \right] \max_j \|\hat{\beta}_j(r)\|_1 \\ & \geq \frac{1}{pN} \left\| \mathbf{A}_k^T \mathbf{y} \right\|_1 - r \end{aligned}$$

By Lemma 3, the positivity assumption on  $\beta^0$ , and the condition  $\sum_{n/[T_n]=t_k} E_n \geq e_{\min}$  it holds under probability greater than  $1 - N \exp(-N(\tau + 1)/2)$  that:

$$\frac{1}{pN} \left\| \mathbf{A}_k^T \mathbf{y} \right\|_1 \geq \mu e_{\min} - 2\sigma\sqrt{\tau+1} - \frac{\|\delta\|_2}{\sqrt{N}}$$

which yields to the following: whenever it holds that

$$\begin{aligned} \mu e_{\min} & > 2\sigma\sqrt{\tau+1} + \frac{\|\delta\|_2}{\sqrt{N}} + r + (2m+1)\alpha \\ & + 2 \left[ \sum_{i>m} \rho(i) \right] \max_j \|\hat{\beta}_j(r)\|_1 \end{aligned}$$

then with probability greater than  $1 - N \exp(-N(\tau + 1)/2)$  it also holds that:

$$\sum_{|j-k| \leq m} \|\hat{\beta}_j(r)\|_1 > (2m+1)\alpha ;$$

the proposition immediately follows.

### E. Proof of Proposition 4

We shall make use of the following other estimators of  $\lambda$  as intermediates:

- 1) the *ideal* estimator  $\hat{\lambda}_{ideal}$  which estimates from the Poissonian sample ; in other words, should the arrival times  $T_k$  and their number  $M$  be known, a standard kernel estimator of the activity would be

$$\hat{\lambda}_{ideal}(t) = \frac{1}{h} \sum_{k=1}^M W\left(\frac{t-T_k}{h}\right) \quad (32)$$

- 2) the *quasi-ideal* estimator  $\hat{\lambda}_{quasi}$  which estimates from the quantized pattern  $P_0$ , but exploiting for each integer in  $P_0$  the number of pulses related to it ; remember that the signal is sampled at times  $\{t_n, n = 0 \dots N-1\}$ , so it is impossible to know exactly where the photon arrivals occurred. However, if we do know how many arrivals occurred between consecutive sampling points, then (32) can be slightly modified to provide good results:

$$\hat{\lambda}_{quasi}(t) = \frac{1}{h} \sum_{k=0}^{N-2} a_k W\left(\frac{t-t_k}{h}\right), \quad (33)$$

where  $a_k \triangleq \#\{n = 1, \dots, M ; t_k = [T_n]\}$  defines the number of real arrivals whose rounding is  $t_k$ .



With the previous definition of the  $a_k$ 's, it is clear that the oracle estimator  $\hat{\lambda}_{oracle}$  defined in (25) can be rewritten as:

$$\begin{aligned}\hat{\lambda}_{oracle}(t) &= \frac{1}{h} \sum_{k \in P_0} W\left(\frac{t - t_k}{h}\right) \\ &= \frac{1}{h} \sum_{k=0}^{N-2} 1(a_k > 0) W\left(\frac{t - t_k}{h}\right)\end{aligned}$$

Table III summarizes the definitions of the different estimates appearing in this proof and the next.

| Name                  | Definition                                                                                        | Remarks                                                                                |
|-----------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Ideal estimator       | $\hat{\lambda}_{ideal}(t) = \frac{1}{h} \sum_{k=1}^M W\left(\frac{t - T_k}{h}\right)$             | The arrival times $T_n$ and their number $M$ are both known                            |
| Quasi-ideal estimator | $\hat{\lambda}_{quasi}(t) = \frac{1}{h} \sum_{k=0}^{N-2} a_k W\left(\frac{t - t_k}{h}\right)$     | Only the sampled arrival times are known, but the actual number of pulses $M$ is known |
| Oracle estimator      | $\hat{\lambda}_{oracle}(t) = \frac{1}{h} \sum_{k \in P_0} W\left(\frac{t - t_k}{h}\right)$        | Best attainable estimator based on the sampled signal and the sparsity pattern $P_0$   |
| NIL estimator         | $\hat{\lambda}_{NIL}(t) = \frac{1}{h} \sum_{i=1}^{\hat{M}} W\left(\frac{t - \hat{T}_i}{h}\right)$ | Proposed estimator                                                                     |

TABLE III

DESCRIPTION OF THE ESTIMATES APPEARING IN THE PAPER

Thus the requested error term is decomposed as a sum of the following three terms:

$$\|\lambda - \hat{\lambda}_{oracle}\|_2 \leq \|\hat{\lambda}_{ideal} - \lambda\|_2 \quad (34)$$

$$+ \|\hat{\lambda}_{quasi} - \hat{\lambda}_{ideal}\|_2 \quad (35)$$

$$+ \|\hat{\lambda}_{oracle} - \hat{\lambda}_{quasi}\|_2 \quad (36)$$

We shall provide now bounds on (34), (35), (36).

1) Using Lemma A.1 [37, Supplementary material] (the conditional expectations there are simply expected values here since the function  $\lambda(t)$  is a fixed one):

$$\begin{aligned}\mathbb{E}[\|\lambda - \hat{\lambda}_{ideal}\|_2] &= \frac{1}{h} \int_{h\ell(W)/2}^{T-h\ell(W)/2} \left( \int_{-\ell(W)/2}^{\ell(W)/2} W^2(r) \lambda(rh + t) dr \right) dt \\ &+ \int_{h\ell(W)/2}^{T-h\ell(W)/2} \left( \int_{-\ell(W)/2}^{\ell(W)/2} W(r) [\lambda(rh + t) - \lambda(t)] dr \right)^2 dt\end{aligned} \quad (37)$$

Concerning the first term in the RHS of (37):

$$\begin{aligned}&\frac{1}{h} \int_{h\ell(W)/2}^{T-h\ell(W)/2} \left( \int_{-\ell(W)/2}^{\ell(W)/2} W^2(r) \lambda(rh + t) dr \right) dt \\ &\leq \frac{1}{h} \left( \int_{-\ell(W)/2}^{\ell(W)/2} W^2(r) dr \right) \int_0^T \lambda(t) dt\end{aligned}$$

Concerning the second term, using simply the bound valid for all  $t$  in the indicated interval:  $|\lambda(rh + t) - \lambda(t)| \leq \|\lambda\|_\infty$ , we obtain therefore

$$\begin{aligned}&\int_{h\ell(W)/2}^{T-h\ell(W)/2} \left( \int_{-\ell(W)/2}^{\ell(W)/2} W(r) [\lambda(rh + t) - \lambda(t)] dr \right)^2 dt \\ &\leq \|\lambda\|_\infty^2 (T - \ell(W)h)\end{aligned}$$

Thus we have proved:

$$\begin{aligned}\mathbb{E}[\|\lambda - \hat{\lambda}_{ideal}\|_2] &\leq \frac{\|W\|_2^2}{h} \int_0^T \lambda(t) dt + \|\lambda\|_\infty^2 (T - \ell(W)h)\end{aligned} \quad (38)$$

2) For every  $j = 1, \dots, M$  let us define  $d_j \triangleq T_j - [T_j]$ . Obviously we do have  $|d_j| \leq \Delta t/2$ . Thus, in order to bound the error made between  $\hat{\lambda}_{ideal}$  and  $\hat{\lambda}_{quasi}$  we write:

$$\begin{aligned}\|\hat{\lambda}_{ideal} - \hat{\lambda}_{quasi}\|_2 &\leq \frac{1}{h} \sum_{k=1}^M \left\| W\left(\frac{t - T_k}{h}\right) - W\left(\frac{t - T_k + d_k}{h}\right) \right\|_2\end{aligned}$$

Now we do have  $\left| W\left(\frac{t - T_k}{h}\right) - W\left(\frac{t - T_k + d_k}{h}\right) \right| \leq \|W'\|_\infty \frac{|d_k|}{h}$  for all  $t$ , and nonzero only when  $|t - T_k| < h \frac{\ell(W)}{2} + |d_k|$ ; thus we can write

$$\begin{aligned}\|\hat{\lambda}_{ideal} - \hat{\lambda}_{quasi}\|_2 &\leq \frac{\Delta t \sqrt{\Delta t + h\ell(W)}}{2h^2} \|W'\|_\infty \cdot M\end{aligned}$$

Therefore we have proved the following:

$$\begin{aligned}\mathbb{E}[\|\hat{\lambda}_{ideal} - \hat{\lambda}_{quasi}\|_2] &\leq \frac{\Delta t \sqrt{\Delta t + h\ell(W)}}{2h^2} \|W'\|_\infty \int_0^T \lambda(t) dt ;\end{aligned}$$

3) We shall now bound in average the error term  $\|\hat{\lambda}_{quasi} - \hat{\lambda}_{oracle}\|_2$ .

$$\begin{aligned}\mathbb{E}[\|\hat{\lambda}_{quasi} - \hat{\lambda}_{oracle}\|_2] &\leq \frac{\|W\|_2}{\sqrt{h}} \sum_{k=0}^{N-2} \mathbb{E}[a_k - \mathbf{1}(a_k > 0)] \\ &= \frac{\|W\|_2}{\sqrt{h}} \sum_{k=0}^{N-2} \left\{ \int_{(k-\frac{1}{2})\Delta t}^{(k+\frac{1}{2})\Delta t} \lambda(u) du + \exp\left(- \int_{(k-\frac{1}{2})\Delta t}^{(k+\frac{1}{2})\Delta t} \lambda(u) du\right) - 1 \right\}\end{aligned}$$

It is easily verified that for any real  $x \in (0; 3)$  the inequality  $x + \exp(-x) - 1 \leq \frac{x^2}{2}$  does hold, hence combined with the in-

equality  $\left( \int_{(k-1/2)\Delta t}^{(k+1/2)\Delta t} \lambda(u) du \right)^2 \leq \|\lambda\|_\infty \Delta t \int_{(k-1/2)\Delta t}^{(k+1/2)\Delta t} \lambda(u) du$   
 we have obtained that:

$$\mathbb{E} \left[ \left\| \hat{\lambda}_{quasi} - \hat{\lambda}_{oracle} \right\|_2 \right] \leq \frac{\|W\|_2}{2\sqrt{h}} \|\lambda\|_\infty \Delta t \int_0^T \lambda(u) du \quad (39)$$

The proof of the proposition is complete by combining all the previous terms.

## REFERENCES

- [1] D.J. Lingensfelter, J.A. Fessler, C.D. Scott, and Zhong He, "Asymptotic source detection performance of gamma-ray imaging systems under model mismatch," *IEEE Transactions on Signal Processing*, vol. 59, no. 11, pp. 5141–5151, Nov 2011.
- [2] N. Dobigeon, J.-Y. Tourneret, and J.D. Scargle, "Joint segmentation of multivariate astronomical time series: Bayesian sampling with a hierarchical model," *IEEE Transactions on Signal Processing*, vol. 55, no. 2, pp. 414–423, Feb 2007.
- [3] J.F. Oliver and M. Rafecas, "Singles-prompts-randoms: Estimation of spurious data rates in pet," in *Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC)*, 2012 IEEE, Oct 2012, pp. 2995–2997.
- [4] G. Tong Zhou, W.R. Schafer, and R.W. Schafer, "A three-state biological point process model and its parameter estimation," *IEEE Transactions on Signal Processing*, vol. 46, no. 10, pp. 2698–2707, Oct 1998.
- [5] Y. Wang, A. Paiva, J. Principe, and J. Sanchez, "Sequential monte carlo point-process estimation of kinematics from neural spiking activity for brain-machine interfaces," *Neural Computation*, vol. 21, no. 10, pp. 2894–2930, Oct 2009.
- [6] Z. Pawlas, L. Klebanov, M. Prokop, and P. Lansky, "Parameters of spike trains observed in a short time window," *Neural Computation*, vol. 20, no. 5, pp. 1325–1343, May 2008.
- [7] T.P. Coleman and Sridevi Sarma, "A computationally efficient method for modeling neural spiking activity with point processes nonparametrically," in *Decision and Control, 2007 46th IEEE Conference on*, Dec 2007, pp. 5800–5805.
- [8] S. Sarma, D. Nguyen, G. Czanner, S. Wirth, M. Wilson, W. Suzuki, and E. Brown, "Computing confidence intervals for point process models," *Neural Computation*, vol. 23, no. 11, pp. 2731–2745, Nov 2011.
- [9] G. F. Knoll, *Radiation Detection and Measurement*, Wiley, 2nd edition, 1989.
- [10] N. Antoniadis and A.O. Hero, "Time-delay estimation for filtered poisson processes using an em-type algorithm," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2112–2123, Aug 1994.
- [11] H. Mino, "Parameter estimation of the intensity process of self-exciting point processes using the em algorithm," *IEEE Transactions on Instrumentation and Measurement*, vol. 50, no. 3, pp. 658–664, Jun 2001.
- [12] A. Jezierska, C. Chaux, J.-C. Pesquet, H. Talbot, and G. Engler, "An em approach for time-variant poisson-gaussian model parameter estimation," *IEEE Transactions on Signal Processing*, vol. 62, no. 1, pp. 17–30, Jan 2014.
- [13] Y. Salimpour, H. Soltanian-Zadeh, and M.D. Abolhassani, "Extended kalman filtering of point process observation," in *Engineering in Medicine and Biology Society (EMBC), 2010 Annual International Conference of the IEEE*, Aug 2010, pp. 6670–6673.
- [14] A. Ergun, R. Barbieri, Uri T. Eden, M.A. Wilson, and E.N. Brown, "Construction of point process adaptive filter algorithms for neural systems using sequential monte carlo methods," *IEEE Transactions on Biomedical Engineering*, vol. 54, no. 3, pp. 419–428, March 2007.
- [15] H. Mino and K. Yana, "Estimation of the density of the filtered poisson impulse process: a parametric approach," *IEEE Transactions on Signal Processing*, vol. 41, no. 4, pp. 1710–1714, Apr 1993.
- [16] A. Rivoira and G.A. Fleury, "A consistent nonparametric spectral estimator for randomly sampled signals," *IEEE Transactions on Signal Processing*, vol. 52, no. 9, pp. 2383–2395, Sept 2004.
- [17] T.P. Coleman and S. Sarma, "Using convex optimization for nonparametric statistical analysis of point processes," in *Information Theory, 2007. ISIT 2007. IEEE International Symposium on*, June 2007, pp. 1476–1480.
- [18] T. Coleman and S. Sarma, "A computationally efficient method for nonparametric modeling of neural spiking activity with point processes," *Neural Computation*, vol. 22, no. 8, pp. 2002–2030, Aug 2010.
- [19] R.E. Sequeira and J.A. Gubner, "Intensity estimation from shot-noise data," *IEEE Transactions on Signal Processing*, vol. 43, no. 6, pp. 1527–1531, Jun 1995.
- [20] R.E. Sequeira and J.A. Gubner, "Blind intensity estimation from shot-noise data," *IEEE Transactions on Signal Processing*, vol. 45, no. 2, pp. 421–433, Feb 1997.
- [21] M. Unser and P.D. Tafti, "Stochastic models for sparse and piecewise-smooth signals," *IEEE Transactions on Signal Processing*, vol. 59, no. 3, pp. 989–1006, March 2011.
- [22] U. S. Kamilov, P. Pad, A. Amini, and M. Unser, "Mmse estimation of sparse lévy processes," *IEEE Transactions on Signal Processing*, vol. 61, no. 1, pp. 137–147, Jan 2013.
- [23] E. J. Candes and Y. Plan, "A Probabilistic and RIPless Theory of Compressed Sensing," *IEEE Transactions on Information Theory*, vol. 57, no. 11, pp. 7235–7254, Nov 2011.
- [24] G. Lecue and S. Mendelson, "Regularization and the Small-Ball Method I: Sparse Recovery," Submitted, 2016.
- [25] Y. Sepulcre, T. Trigano, and Y. Ritov, "Sparse regression algorithm for activity estimation in gamma spectrometry," *IEEE Transactions on Signal Processing*, vol. 61, no. 17, pp. 4347–4359, Sept 2013.
- [26] T. Trigano, Y. Sepulcre, M. Roitman, and U. Aferiat, "On nonhomogeneous activity estimation in gamma spectrometry using sparse signal representation," in *Statistical Signal Processing Workshop (SSP), 2011 IEEE*, June 2011, pp. 649–652.
- [27] C. Giraud, S. Huet, and N. Verzelen, "High-Dimensional Regression with Unknown Variance," *Statistical Science*, vol. 27, no. 4, pp. 500–518, 2012.
- [28] S. Chretien and S. Darses, "Sparse Recovery with Unknown Variance: a LASSO-type Approach," *IEEE Transactions on Information Theory*, vol. 60, no. 7, pp. 3970–3988, 2014.
- [29] D.J. Daley and D. Vere-Jones, *An Introduction to the Theory of Point Processes - Volume I: Elementary Theory and Methods*, Springer Series in Statistics. Springer-Verlag, 2 edition, 2003.
- [30] P. A. W. Lewis and G. S. Shedler, "Statistical analysis of non-stationary series of events in a data base system," *IBM J. Res. Dev.*, vol. 20, no. 5, pp. 465–482, 1976.
- [31] E. Barat, T. Dautremer, and T. Trigano, "Nonparametric Bayesian Estimation of Censored Counter Intensity from the Indicator Data," in *26th International Workshop on Bayesian Inference and Maximum Entropy Methods in Science and Engineering*, 2006.
- [32] R. Tibshirani, "Regression shrinkage and selection via the LASSO," *Journal of the Royal Statistical Society*, vol. 58, no. 1, pp. 267–288, 1996.
- [33] B. Efron, T. Hastie, I. Johnstone, and R. Tibshirani, "Least angle regression," *Annals of Statistics*, vol. 32, no. 2, pp. 407–499, 2004.
- [34] P. Diggle, "A Kernel Method for Smoothing Point Process Data," *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, vol. 34, no. 2, pp. 138–147, 1985.
- [35] M. C. Jones, J. S. Marron, and S. J. Sheather, "A brief survey of bandwidth selection for density estimation," *Journal of the American Statistical Association*, vol. 91, no. 433, pp. 401–407, 1996.
- [36] S. Mallat, *A Wavelet Tour in Signal Processing – The Sparse Way*, Academic Press, 2nd edition, 2009.
- [37] T. Zhang and S. C. Kou, "Nonparametric inference of doubly stochastic poisson process data via the kernel method," *Ann. Appl. Stat.*, vol. 4, no. 4, pp. 1913–1941, 12 2010.



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